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RahZil

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Abstract

The potential of the Pakistan tourism industry to attract people from all across the world and generate worthwhile revenue for the locals is not an unknown fact. This is made possible because of the diverse range of attractions that lie from the North to the South of Pakistan, consisting of not only lush green mountains, deep scenic valleys and crystal clear lakes, but also historic sites and beautiful monuments. Unfortunately, the huge inflow of tourists poses certain challenges that affect the quality of travel planning and management. To counter these, RahZil has been developed, revolutionizing the way we travel. As a single platform providing aid in various aspects of tourism, our web application comes forth as a comprehensive travel assistance package. Different features of RahZil prove to be of value to different users including tourists, travel agents, bus managers and hotel administrators. The process of reserving buses, booking travel agents, finding accommodations, creating personalized travel itineraries, accessing real-time weather updates and managing budgets is simplified for the tourists. For service providers, RahZil introduces tools that allow them to efficiently upload travel packages and bus schedules and manage room availability, pricing and bookings. RahZil's user-friendly interface and robust functionalities contribute to its potential to transform the tourism landscape in Pakistan. By addressing the challenges faced by travelers and industry stakeholders, RahZil aims to enhance the overall tourism experience, promote sustainable tourism practices, and contribute significantly to the economic growth of the region. It undoubtedly holds the potential to make travel experiences in Pakistan unparalleled in terms of quality.

Executive Summary

With an increase in the tourist inflow, Pakistan has been under a lot of pressure to maintain quality in the tourism industry. RahZil, our web application, offers a remedy to the resulting issues. An increase in tourist visits has made it clear that organizing travel, keeping track of reservations, and facilitating effective communication between all parties involved are difficult tasks.

RahZil promises to be a game-changing web application that unites all aspects of tourism into a single user platform with the aim to successfully address these problems. RahZil's main objective is to make tourism easier for everyone involved—tourists, travel agencies, bus company managers, and hotel administrators.

RahZil wants to provide a seamless system that simplifies every part of travel by fusing technology use with Pakistan's ability to attract tourists with the diverse range of tourist attractions. Our application offers a range of tools that enable travelers to plan their trips, book rooms, buses and travel agents, access weather forecast information in real time, and keep a check on their budget.

At the same time, managers of bus companies, hotel administrators, and travel agencies can effectively list their services for users to acquire while engaging with them to increase the overall efficiency of their business. Other than promoting ease, RahZil supports sustainable development goals. It is a driving force for positive change in Pakistan's tourism industry because it supports responsible tourist behavior and resource consumption and develops local economies. RahZil has strategic advantages as it works on centralizing the tourist landscape, encouraging connectivity and significantly improving user experience.

It also empowers travelers and businesses through the use of technology, creating better synergy that leads to a stress-free and unforgettable travel experience. RahZil's functionalities address the problems hindering the growth of tourism sector in Pakistan by taking advantage of the power of digital inventions. It is bringing in a new age of simple, sustainable, and memorable travel experiences by becoming a catalyst for revolutionary transformation.

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Chapter 1 Introduction

RahZil intends to bring about change within the Pakistani tourism industry. In a country with a wide variety of attractions, from scenic mountains to pristine landscapes and archaeological wonders, tourism presents opportunities as well as challenges. To accelerate trends in tourism, we need to overcome the challenges presented to us. RahZil aims to facilitate travel agents, bus companies and hotels along with visitors to make each experience worthwhile. By integrating multiple components into a single web application, RahZil has become the modern way to travel.

1.1 Purpose of this Document

The purpose of this document is to provide a comprehensive and systematic overview of RahZil to users, developers and stakeholders. It clarifies its vision, goals, objectives and technical requirements, and lays the foundation for any plan to transform Pakistan's tourism industry. This document's clearly defined specifications for RahZil indicate that it will serve as a main point of reference for the duration of the project for making informed decisions and devising strategic plans.

1.2 Intended Audience

The audience for this document includes RahZil's developers, designers, and project managers, as well as its users, such as travelers, hotel managers, bus managers, and travel agencies. The document contains several sections, providing information to different audience groups that they may find useful over the course of development and use of RahZil.

1.3 Definitions, Acronyms, and Abbreviations

The following is a list of abbreviations used throughout the document along with their definitions:

RahZil: 'Rahbar' (guide) and 'Manzil' (destination).

RahZil: '**Rahbar**' (guide) and '**Manzil**' (destination).

UI: User Interface

UX: User Experience

API: Application Programming Interface

GPS: Global Positioning System

CPEC: China-Pakistan Economic Corridor

1.4 Conclusion

RahZil's focus on a transformative journey within the tourism industry in Pakistan is highlighted by its goals, vision, technical requirements, and objectives as specified in this document. All aspects of RahZil's design, development, and use have been laid out in detail in the following chapters to enable efficient collaboration, understanding and implementation. It is important to note that development of such an application is pivotal for the success of tourism in Pakistan.

Chapter 2 Project Vision

RahZil's driving aim is to simplify the many layers of travel planning and administration in order to revolutionize tourism. It aspires to empower tourism with a one-click away platform in a country full of natural and historical attractions. Passengers may use RahZil to book buses, interact with travel agents, book lodgings, keep up with their schedules and finances, and receive real-time weather predictions. It provides a platform for flexible scheduling and management for transport operators, bus company managers, and hotel managers, in order to enhance the passenger experience. By allowing the users to upload and acquire services within a unified digital environment, RahZil leads to greater efficiency and better collaboration.

2.1 Problem Domain Overview

Travel management is a major challenge to Pakistan's expanding tourism sector. The nation is seeing an increase in visitors, but standard methods of organizing, booking, and planning have become inefficient. Now, RahZil stands out as a comprehensive solution that connects travelers, tour operators, transport companies, and hotels. It aims to overcome the difficulties in managing transport and to offer a common ground on which different stakeholders may work together productively.

2.2 Problem Statement

The division of Pakistan's tourist sector is the root of the current issue. While companies struggle with difficulties in managing and reserving services, travelers face difficulties in locating trustworthy providers and organizing their journeys. By centralizing these services, RahZil offers a better user experience, solving issues by changing the way people travel around Pakistan.

2.3 Problem Elaboration

Unreliable reservation systems, differing management quality, and a lack of a single platform are some of the problems faced by the tourist sector in Pakistan. Travelers frequently have trouble managing different aspects of their vacation, which leads to a bad experience. Service providers work hard to properly process bookings and reach their target audience at the same time. RahZil digs deeper into these issues while providing flexible ways that cover the gap between customers and service providers, creating an enjoyable and stress-free tourism experience for all.

2.4 Goals and Objectives

Improving and simplifying the tourism experience in Pakistan is the primary objective of RahZil. It aims to make travel planning simple for customers and to give bus, travel companies, and hotel operators a platform through which they can manage their businesses more effectively. The objectives include:

- Designing an interface that's easy to use
- Streamlining booking processes
- Ensuring real-time travel aid
- Enabling ease for tourists

RahZil aims to not only resolve existing issues but also elevate the overall tourism landscape in Pakistan.

2.5 Project Scope

Rahzil's aim is to improve Pakistan's entire tourism landscape as well as to tackle existing difficulties. Its influence is felt throughout the travel sector. It offers tourist related features as well as those related to service providers. It allows travelers to book buses, travel agents and accommodations, make personalized travel plans, obtain weather forecasts information and manage their finances along with itineraries. At the same time, it allows the hotels, travel agents and bus companies to upload their rooms, travel packages and buses, respectively, for tourists to book. From the beginning to the end of the travel experience, RahZil offers facilities that promote ease and growth.

2.6 Sustainable Development Goal (SDG)

RahZil aligns significantly with Goal 8 of Sustainable Development Goals: Decent Work and Economic Growth. By facilitating business, RahZil encourages economic growth in the tourism sector, creates job opportunities and encourages sustainable development practices. Its influence extends to local economies. As the local businesses and communities are empowered, the broader goal of quality jobs and economic growth within Pakistan is supported.

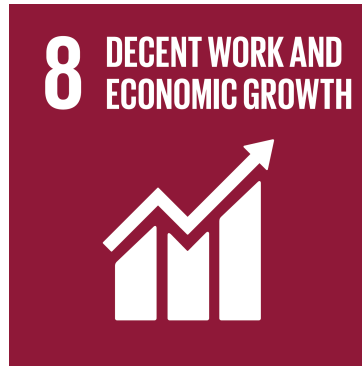


Figure 2.1: Decent Work and Economic Growth

By facilitating business, Rahzil provides economic growth via empowering local businesses

Chapter 3 Literature Review / Related Work

In the following section of this literature review, we will immerse ourselves in the intricate landscape of related research and existing solutions within the vibrant travel and tourism sector. While an array of platforms and services caters to the diverse facets of the travel industry, our primary goal is to introduce RahZil as a pioneering, all-encompassing solution. This comprehensive review will delve into analogous web applications and services, meticulously dissecting their features, limitations, and areas where RahZil seeks to assert itself as a trailblazing force, poised to redefine the very essence of travel and tourism.

3.1 Definitions, Acronyms, and Abbreviations

There are the following important definitions, the acronyms and abbreviations used in this section.

B&B: Bed and Breakfast

TDL :To-do List

WF: Weather Forecast

BM: Budget Monitoring

3.2 Detailed Literature Review

3.2.1 Hotels Application

Hotels.com is a website that offers a rewards program for guests aged 18 and above who sign up with a valid email address. Booking and staying for one night at a qualifying property will earn users one stamp. Collect 10 stamps, and users will get one reward night to redeem at any eligible property. The website also offers a range of accommodation options, including hotels, apartments, and vacation rentals[1].

3.2.1.1 Summary

Online or over-the-phone hotel reservations are available through the website Hotels.com. The company has 85 websites with over 325,000 hotels spread over 19,000 locations and 34 languages. Apart from hotels, the website provides plenty of lodging choices, such as condominiums, flats, and holiday rentals. Vacation rental properties are a dynamically different way to travel compared to hotel rooms. When you search for a vacation rental on Hotels.com, you get exclusive access to vacation home rentals all over the world. Choosing a vacation rental, such as a cottage in the woods or a city condo, will make your trip comfortable and seem like home away from home. The website also provides a search engine

that allows you to compare prices across multiple hotel booking websites. This is handy when you want to find the best deal on a hotel room.

3.2.1.2 Critical Analysis

Hotels.com distinguishes itself in the online accommodation booking sphere through its enticing rewards program, enabling users to earn stamps and unlock complimentary nights, fostering customer loyalty. Operating across a vast network of 85 websites in 34 languages, it offers access to over 325,000 hotels globally, catering to diverse preferences and travel destinations.

Nonetheless, there's room for improvement. Providing more comprehensive property descriptions and an extensive repository of guest reviews would empower travelers to make more informed decisions. Additionally, personalized travel recommendations tailored to individual preferences could elevate the user experience and encourage exploration of new destinations.

In a highly competitive landscape, Hotels.com must continually innovate to provide enduring value. Enhancing property information, nurturing personalized recommendations, and ensuring responsive customer support can help solidify its position as a trusted partner in travel and accommodation booking.

3.2.1.3 Relationship to the proposed research work

Hotels.com, while sharing similarities with our proposed project, primarily focuses on the online booking and reservation aspects of the travel and tourism industry. In our research, we aim to explore and enhance the user experience within the travel domain by introducing features that encompass not only accommodation booking but also broader travel planning and management. While Hotels.com excels in offering a comprehensive array of lodging options and a rewards program, our project seeks to go beyond by integrating features like bus reservations, travel agent bookings, TDL management, WF, BM, and the provision for hotel and bus managers to list their services. Our vision extends towards a holistic travel and tourism solution that caters to the diverse needs of travelers, encompassing more than just accommodation bookings.

3.2.2 Busbud Application

Busbud.com functions as a worldwide marketplace, streamlining the search for and comparison of reasonably priced bus and train tickets. Its wide linguistic support makes its services accessible to people with a variety of backgrounds. In addition to interacting with Booking.com to find housing, Busbud.com provides travellers with tips on how to save money.

3.2.2.1 Summary

Busbud.com is a service that assists travellers in finding low-cost bus and train tickets all around the world. The website gives information on bus travel alternatives and allows users to compare and purchase tickets on over 2 million bus routes in 80 countries. Busbud.com collaborates with over 4,500 bus operators to provide a diverse selection of options. Booking.com allows users to conveniently organise their journeys by looking for hotel rooms. The application supports many languages and currencies, allowing users to use the language purposefully and convert their currency. The Busbud website is simple to use, allowing users to quickly discover and book their favourite bus. The service also keeps users up to speed on their schedules, assisting them on their way. BusBud.com also provides customer service 24/7.

3.2.2.2 Critical Analysis

If we talk about the strengths of busbud.com, it allows us to compare the bus ticket prices in 80 countries on more than 2 million bus routes. It also gives users early booking and advices them to not travel during peak hours so users can save their money. Its website is also very simple and easy to understand so users can book buses easily. Busbud also partners up with local companies to save their environment by eliminating carbon pollution. By allowing people to directly share their travel experiences on the website, Busbud.com could be improved. It is now difficult for users to provide comments, which makes it more difficult for other individuals to choose their vacation itinerary. It would also be beneficial if they provided more specific information about the routes, such as how long they take, how frequently they stop, and what amenities the buses offer.

3.2.2.3 Relationship to the proposed research work

Busbud.com aligns with our proposed project, focusing on online booking and reservation aspects within the travel and tourism industry, particularly in the bus and train ticket sector. While Busbud.com excels in this niche, our project, TravelEase, aims for a broader scope by incorporating features like hotel and travel agent bookings, TDL, WF, BM, and services listings for hotel and bus managers (SL-HBM). Our vision is to provide a comprehensive travel and tourism platform catering to diverse traveler needs beyond ticket reservations.

3.2.3 Trafalgar Application

Trafalgar.com is a website that offers tour packages and vacations for travelers who want to explore the world with confidence. The website provides deals on tours departing soon, last-minute deals, and

vacations under 14 days[2].

3.2.3.1 Summary

Trafalgar is an online travel portal designed to help tourists look for vacations and routes throughout the globe. The website features last-minute offers, tours departing soon, and vacation packages shorter than 14 days. Trafalgar provides domestic tours, budget trips, and a Global Tour Rewards savings account. It offers unique experiences, including local culture and insider secrets that are only available to those who travel to more than 80 countries spread across seven continents. Accessible and sustainable transport is a priority for Trafalgar. The platform collaborates with community groups to encourage eco-friendly travel choices and lower carbon emissions.

3.2.3.2 Critical Analysis

Travelers can quickly search for and reserve tours on the user-friendly Trafalgar website. Users can be sure they have up-to-date information about their travel plans with the platform's real-time updates on tour schedules and delays. The lack of an online review system, however, is a significant flaw that makes it difficult for prospective customers to make wise choices when purchasing their airline tickets. More detailed route information, such as the length of the journey, the number of stops, and the amenities offered on board, should be provided.

3.2.3.3 Relationship to the proposed research work

While both RahZil and Trafalgar.com are in the tourist industry, their areas of expertise are distinct. Trafalgar.com specializes in tour packages and vacations, emphasizing curated experiences and sustainability[2]. On the other hand, RahZil aims to be a comprehensive web application, offering a wider range of travel services, including bus reservations, travel agent bookings, hotel accommodations, and more. While Trafalgar.com aligns with specific aspects of our RRW related to tour packages and sustainability, RahZil's ambition extends to provide a complete travel solution for a variety of traveler needs beyond tour bookings.

3.2.4 Viator Application

Viator, headquartered in San Francisco, is a global company that helps you book tours and attractions. More than 300,000 city excursions, day trips, tourist trips, and other options are available from them. Viator collaborates with local tour guides, linking them with travelers planning trips to their area[3].

3.2.4.1 Summary

One of the top travel websites, TripAdvisor, is the parent company of Viator. Viator links local tour guides with tourists who are organising trips to their region. It makes it easier for small, local businesses to find clients without requiring them to engage in intensive digital marketing. Travellers can use the website of Viator to look for tours and activities according to their preferences, dates, and destination. A thorough description is included for every tour or activity, giving guests important details about what to expect from the tour, including the sights, sounds, and locations of the tour stops during the day. In addition, users can look at pictures and videos from the trip as well as read reviews written by other tourists who have done the same tour. Additionally, Viator provides a low-price guarantee, meaning that customers will receive a refund if, within three days of making their reservation or while travelling, they discover the same tour or activity for less. Furthermore, consumers may cancel most reservations on Viator for a complete refund up to 24 hours prior to the experience's start date because to the company's cancellation policy.

3.2.4.2 Critical Analysis

Travelers often choose Viator because of its many advantages. It collaborates with regional guides and small companies to provide a diverse array of tours and activities throughout the globe. Travelers may quickly locate trips based on their tastes thanks to the website's user-friendly design. Extensive explanations of every tour offer insightful insights into what visitors might anticipate. With its flexible cancellation rules and low-price guarantee, Viator also offers peace of mind.

But there are some weaknesses to take into account. There could be differences in quality of trips offered by Viator because it depends on small enterprises and local guides. Furthermore, the low-price promise may not always apply, as it is conditional on users discovering the identical trip at less money within three days of reserving or during their stay at the destination.

3.2.4.3 Relationship to the proposed research work

Users can use both RahZil and Viator to plan their trips. However, they have different focuses. RahZil is a comprehensive web application that offers a wider range of travel services, including bus reservations, travel agent bookings, hotel accommodations, and more. However, Viator is a global tour and attraction booking firm. With more than 300,000 alternatives, they provide a wide variety, including sightseeing tours, city tours, day trips, and other trips in locations all over the world. While there is some overlap between RahZil's features and Viator's services, they are not directly comparable as they serve different purposes.

3.2.5 Priceline Application

Priceline.com is an online travel service that offers low-cost plane tickets and accommodations[4].

3.2.5.1 Summary

Priceline.com is a travel website based in Norwalk, Connecticut. Like other websites like Booking.com and Kayak.com, it is owned by Booking Holdings.. Priceline operates globally, offering over 2 million hotels and working with 400 airlines and 300,000 hotels in more than 200 countries.

Users can discover great deals on flights and hotels on the Priceline website. They used to feature "Name Your Own Price," allowing users to bid on hotel rooms and flights. Now, they offer Express Deals, where users see the price and location without knowing the hotel name[4].

Beyond hotels, Priceline also offers discounted cruises, tours, and attractions. Their tool, Pricebreakers, for hotel bookings claims up to 50% off retail prices. It's similar to Express Deal but displays the names of three hotels, revealing the actual one after payment.

For users who need to change plans, Priceline has a flexible policy. Most bookings can be canceled for a full refund up to 24 hours before the experience starts.

3.2.5.2 Critical Analysis

One of Priceline's strengths is that it partners with 300,000 hotels and 400 airlines to offer a vast array of hotels and travel services worldwide. Users may cancel most reservations for a complete refund up to 24 hours prior to the commencement of the event thanks to their accommodating booking procedures. In addition, they provide a low-price guarantee and a feature called Pricebreakers for Hotel Bookings that allows users to save up to 50% on retail costs.

But Priceline is not without its problems. The majority of its activities are centred on the US, and it is simple for new competitors to enter the market. There is a strong dependence on travel agencies and middlemen. Additionally, since 2016, the once-favorite "Name Your Own Price" function has not been accessible.

3.2.5.3 Relationship to the proposed research work

RahZil aims to be a comprehensive web application that offers a wider range of travel services, including bus reservations, travel agent bookings, hotel accommodations, and more. In contrast, Priceline is an online travel agency that offers discounts on accommodations and flight reservations. The company facilitates the provision of trips to its consumers by its providers.. Priceline operates in more than 200 countries and territories globally and provides over 2 million hotels. More than 400 airlines and 300,000

hotels are its partners. While there is some overlap between RahZil's features and Priceline's services, they are not directly comparable as they serve different purposes.

3.2.6 CheckMyBus Application

On the website CheckMyBus, customers may compare rates, check bus timetables, and purchase bus tickets online. They provide many opportunities and millions of routes all over the world. In addition to having simple access to exact locations, departure and arrival times, and total journey times, users may obtain the cheapest bus ticket prices[5].

3.2.6.1 Summary

CheckMyBus offers a range of choices for millions of routes throughout the globe. All users have to do is input their trip dates and destination. After that, they can look up every bus choice and use filters like company or bus type. Every bus route has information about its destinations, hours of departure and arrival, length, and cost. Along with ratings for the bus operator, users may read reviews written by other travellers who have taken the same route[5].

CheckMyBus offers a guarantee of low prices. Within 24 hours of making their reservation, customers will receive a refund if they discover the identical bus route for less money. CheckMyBus serves over 80 countries, over 100,000 cities, and over 135,000 bus stops globally by partnering with thousands of bus providers.

3.2.6.2 Critical Analysis

CheckMyBus boasts several strengths that enhance its appeal to users. These strengths encompass its extensive global coverage of bus routes, comprehensive descriptions of each route, and a reassuring low-price guarantee, instilling confidence in users during the booking process.

However, there are notable weaknesses worth addressing. CheckMyBus heavily relies on third-party bus companies for schedule and price information, potentially resulting in variations in data quality across different routes. Furthermore, its low-price guarantee may not consistently apply as it hinges on users discovering a cheaper fare for the same route within a tight 24-hour window after booking[5].

To enhance its offering, CheckMyBus could benefit from reducing its dependence on third-party data sources and exploring options to promote eco-friendly travel, thereby appealing to environmentally conscious travelers and aligning with evolving travel trends.

3.2.6.3 Relationship to the proposed research work

RahZil aspires to be an all-encompassing travel platform, offering features like bus reservations, travel agent bookings, hotel accommodations, and more. In contrast, CheckMyBus specializes in streamlining global bus travel options. RahZil aims to simplify travel planning, catering to a wide range of traveler needs, including to-do lists, weather forecasts, and budget monitoring. It also connects hotel and bus managers to list their services. On the other hand, CheckMyBus focuses on providing detailed information about bus routes worldwide, emphasizing user reviews and a low-price guarantee. While both platforms operate in the travel sector, they serve different niches, making direct comparisons challenging.

3.3 Literature Review Summary Table

Table 3.1: Comparison of Different Apps

The comparison between 7 apps of various features

Feature	Hotels Applica- tion	BusBud	Trafalgar	Viator	Priceline	Check MyBus	RahZil
Weather Forecast							✓
Reserve Buses		✓				✓	✓
Check Budget							✓
Book Travel Agents			✓	✓			✓
Book Hotel Rooms	✓				✓		✓
Maintain To-Do List							✓

3.4 Conclusion

In this analysis, we look at numerous applications in the travel and tourist business. Our suggested solution, RahZil, intends to provide a comprehensive travel platform that is separate from existing alternatives. We examine platforms such as Hotels.com, Busbud, Trafalgar, Viator, and Priceline, highlighting their advantages and disadvantages. While these platforms specialise in some areas, RahZil anticipates a larger variety of services such as bus reservations, travel agency bookings, hotel rooms, to-do lists,

weather predictions, and budget tracking. This comprehensive strategy distinguishes RahZil, offering travellers an all-encompassing answer to their different travel and tourist demands.

Chapter 4 Software Requirement Specifications

The Software Requirements Specifications (SRS) is the technical guide for RahZil, defining its functional and non-functional requirements. It provides guidance for the development team to design a user-friendly and robust travel companion.

4.1 List of Features

RahZil aims to revolutionize the travel and tourism industry by providing the following features along with basic signup/login functionality as a web application:

- Users can reserve buses
- Users can book travel agents
- Users can book hotel rooms
- Users can maintain a to-do list
- Users have access to weather forecast information
- Users can keep a check on their budget
- Hotel managers can put a room up for rent
- Bus managers can update user about bus availability and schedule
- Travel agents can provide users with travel packages

4.2 Functional Requirements

Along with basic signup and login functionalities, RahZil provides the following features as a web application:

- Users can reserve buses
- Users can book travel agents
- Users can book hotel rooms
- Users can maintain a to-do list
- Users have access to weather forecast information
- Users can keep a check on their budget

- Hotel managers can put a room up for rent
- Bus managers can update user about bus availability and schedule

4.3 Quality Attributes

RahZil seeks to become a secure, reliable and user-friendly application. It fulfills the demands of its users in a way that the following quality attributes stand out clearly, laying the foundation for defining the non-functional requirements.

- Reliability
- Security
- Usability
- Correctness
- Flexibility
- Scalability

4.4 Non-Functional Requirements

- The system shall operate reliably without unexpected failures, ensuring users can access the RahZil platform at all times.
- The system shall maintain data integrity, preventing data loss or corruption during transactions and interactions.
- The system shall implement proper access controls and authorization mechanisms, ensuring that users can only access data and features relevant to their roles.
- The system shall have an intuitive and user-friendly interface, facilitating easy navigation for both tourists and service providers to perform tasks efficiently.
- The system shall provide clear and concise information, guiding users through the booking process and other functionalities.
- The system shall provide accurate and up-to-date information regarding bus availability, hotel bookings, travel packages, and other services.
- The system shall validate user inputs to ensure that data entered is correct and valid, preventing errors and inconsistencies in the system.

- The system shall support various payment methods and integrate with external APIs for services such as weather forecasts.
- The system shall be scalable to handle a growing number of users, transactions, and data volumes without compromising performance.

4.5 Assumptions

It is believed that the user will have an internet connection. They will not be using outdated web browsers, but the ones that enable efficient interaction between the user and web application. Furthermore, the user will have the basic knowledge of navigating through computers and accessing our web application. RahZil also depends on the availability of third party services, such as weather APIs. Hence, it is assumed that reliable information will be provided by them. Lastly, we assume that the service providers will be willing to list their services using our application, helping them in reaching a more diverse market. These assumptions underpin the functionality and usability of RahZil, guiding the development team in creating a robust and user-friendly travel utility application.

4.6 Use Cases

4.6.1 Sign Up

Name	Sample Use Case Name Here		
Actors	Client, Customer		
Summary	The user shall provide their name, phone number, email, and password on the signup form and after successful verification, redirect the user to the sign in page.		
Pre-Conditions	The user must already have a Gmail, yahoo, or Hotmail account to register in app database.		
Post-Conditions	The user is successfully registered and would be redirected to the sign in page.		
Special Requirements	None		
Basic Flow			
Actor Action		System Response	
1	The user opens the signup page.	2	The signup page is displayed asking for username, email, phone number and password.
2	The user enters valid username, email, phone-number, and password.	4	The system verifies the email and password, establishes a session for the user and redirects the user to the home page.
Alternative Flow			
3	The user enters invalid username, phone number, email, or password.	4-A	The system responds with an error message: Incorrect username, phone number, email or password entered.

4.6.2 Sign in

Name	Login		
Actors	Customer, Client		
Summary	The user shall provide their email and password on the login form and after successful verification, redirect the user to the home page.		
Pre-Conditions	The user must be in the database records either added by any of the authorized users or added manually by a developer. The user must not already be logged in.		
Post-Conditions	The user’s session is successfully established and shall be redirected to the home page.		
Special Requirements	None		
Basic Flow			
Actor Action		System Response	
1	The user opens the login page.	2	The login page is displayed asking for email and password.
2	The user enters valid email and password.	4	The system verifies the email and password, establishes a session for the user and redirects the user to the home page.
Alternative Flow			
3	The user enters invalid email or password.	4-A	The system responds with an error message: Incorrect email or password entered.

4.6.3 Sign Out

Name	Sign Out		
Actors	Client, Customer		
Summary	The user presses the sign-out button, which redirects the user to the sign-in page of the app. The user’s profile is signed out simultaneously by the system.		
Pre-Conditions	The user must be in the database records either added by any of the authorized users or added manually by a developer. The user must already be logged in.		
Post-Conditions	The user’s session is terminated, and the user is redirected to the sign-in page.		
Special Requirements	None		
Basic Flow			
Actor Action		System Response	
1	The user presses the sign-out button.	2	The user is logged out by the system, and the login page is displayed asking for email and password.
Alternative Flow			
None		None	

4.6.4 Reserve Transport

Name	Reserve Transport		
Actors	Customer, Client		
Summary	This use case describes the process of a user reserving transport through the system.		
Pre-Conditions	The user is logged into the system.		
Post-Conditions	1. The transport reservation is confirmed. 2. A confirmation message or email is sent to the user.		
Special Requirements	1. The system should display available transport options based on user preferences. 2. Users should be able to specify the date, time, and other preferences for their transport reservation.		
Basic Flow			
Actor Action		System Response	
1	User selects the "Reserve Transport" option.	2	The system presents available transport options.
3	User selects a preferred transport option and specifies details such as date and time.	4	The system validates the information and checks for availability.
5	User confirms the reservation.	6	The system reserves the chosen transport.
Alternative Flow			
None		None	

4.6.5 Book Hotel Rooms

Name	Book Hotel Rooms		
Actors	Customer, Client		
Summary	This use case describes the process of a user booking hotel rooms through the system.		
Pre-Conditions	The user is logged into the system.		
Post-Conditions	The hotel room reservation is confirmed.		
Special Requirements	• The system should display available hotels and rooms based on user preferences. • Users should be able to specify the check-in/out dates, room type, and other preferences.		
Basic Flow			
Actor Action		System Response	
1	User selects the "Book Hotel Rooms" option.	2	The system presents available hotels and rooms.
3	User selects a preferred hotel and room, specifying check-in/out dates and preferences.	4	The system validates the information and checks for room availability.
5	User confirms the room reservation.	6	The system books the chosen hotel room.
Alternative Flow			
None		None	

4.6.6 Book Travel Agents

Name	Book Travel Agents		
Actors	Client, Customer		
Summary	This use case describes the process of a user booking travel agents through the system.		
Pre-Conditions	The user is logged into the system.		
Post-Conditions	• The travel agent reservation is confirmed. • A confirmation message or email is sent to the user.		
Special Requirements	• The system should display available travel agents based on user preferences. • Users should be able to specify the type of travel services needed.		
Basic Flow			
Actor Action		System Response	
1	User selects the "Book Travel Agents" option.	2	The system presents available travel agents and their services.
3	User selects a preferred travel agent and specifies the required services.	4	The system validates the information and checks for agent availability.
5	User confirms the travel agent reservation.	6	The system books the chosen travel agent.
Alternative Flow			
None		None	

4.6.7 Maintain a To-Do List

Name		Maintain a To-Do List	
Actors		Customer	
Summary		This use case describes the process of a user maintaining a to-do list through the system.	
Pre-Conditions		The user is logged into the system.	
Post-Conditions		The to-do list is updated with new tasks or modifications.	
Special Requirements		• The system should provide options to add, edit, and delete tasks from the to-do list. • Users should be able to categorize tasks and set due dates.	
Basic Flow			
Actor Action		System Response	
1	User selects the "Maintain To-Do List" option.	2	The system displays the user's current to-do list.
3	User adds, edits, or deletes tasks from the to-do list.	4	The system updates the to-do list accordingly.
Alternative Flow			
None		None	

4.6.8 Google Map

Name	Google Map		
Actors	Customer		
Summary	The user can see his/her location and route of their plan.		
Pre-Conditions	• The user should be logged in. • The user should have selected a location.		
Post-Conditions	The user can view the route to their planned location.		
Special Requirements	• The system should provide real-time or up-to-date mapping data. • Users should be able to input specific addresses, landmarks, or coordinates to view on the map.		
Basic Flow			
Actor Action		System Response	
1	User selects the "View Map" or similar option.	2	The system opens the map interface.
3	User inputs the desired route or location details.	4	The system processes the user's input and displays the requested route or location on the map.
5	User interacts with the map interface, zooming in, panning, or exploring additional details.	6	The system provides a dynamic and responsive map interface, allowing the user to explore the route or location.
Alternative Flow			
3	If the user inputs invalid or incomplete location information.	4-A	• The system displays an error message. • The user is prompted to provide valid location details.

4.7 Hardware and Software Requirements

The following sections highlight the hardware and software requirements, which if met, will make sure that the application efficiently provides ease to its users and makes their tourism experience rewarding by dispensing to them many different services.

4.7.1 Hardware Requirements

RahZil's hardware requirements have been kept to a minimum to enable this application to be accessed across multiple devices such as laptops, desktop, tablets and smartphones. To be run on devices with moderate processing power and memory, RahZil has been designed to be lightweight. It requires the users to have a stable internet connection to use the application uninterruptedly. Because of the responsiveness of our application, users can run it on different screen sizes and resolutions.

4.7.2 Software Requirements

RahZil requires that the user run it on modern and up-to-date web browsers for a seamless experience and enhanced security. Supported browsers include Google Chrome, Mozilla Firefox, Safari, Opera and Microsoft Edge. Since RahZil is designed using JavaScript to cope with its dynamic nature, JavaScript must be enabled in the user's web browser. The platform relies on secure HTTPS protocols to safeguard data transmission, ensuring encryption and data integrity. Users are encouraged to enable cookies and cache functionalities in their browsers to enhance the performance and user experience of RahZil.

4.8 Graphical User Interface

Following is the GUI of the different screens of this project.5.7.

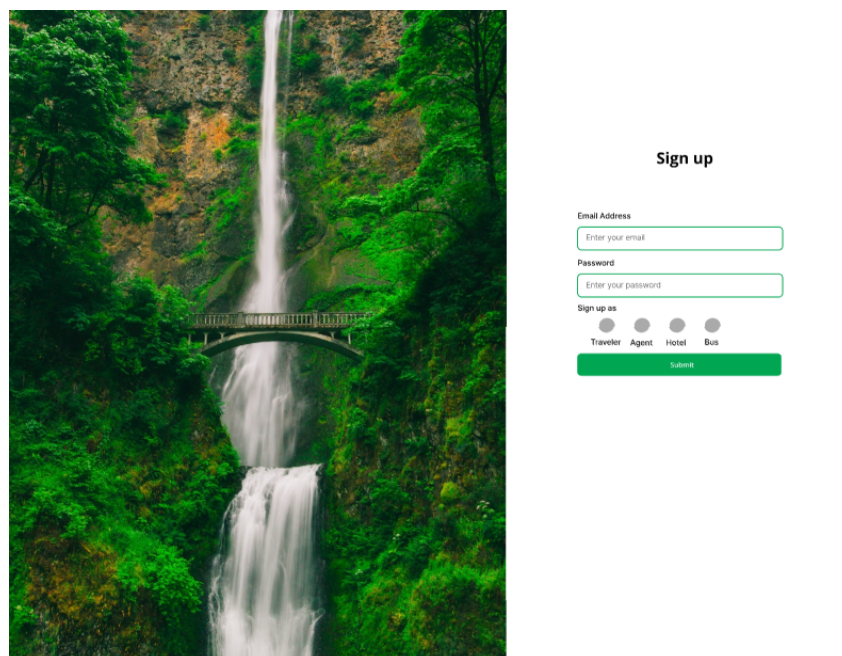
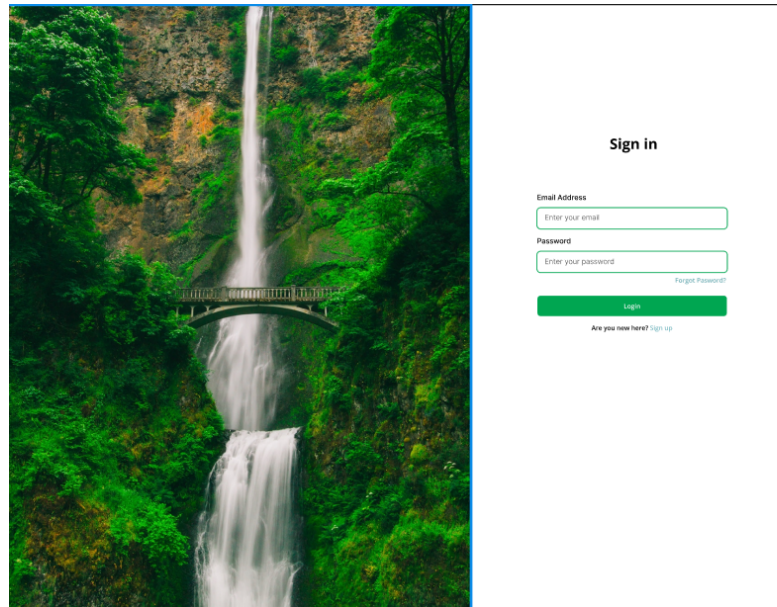
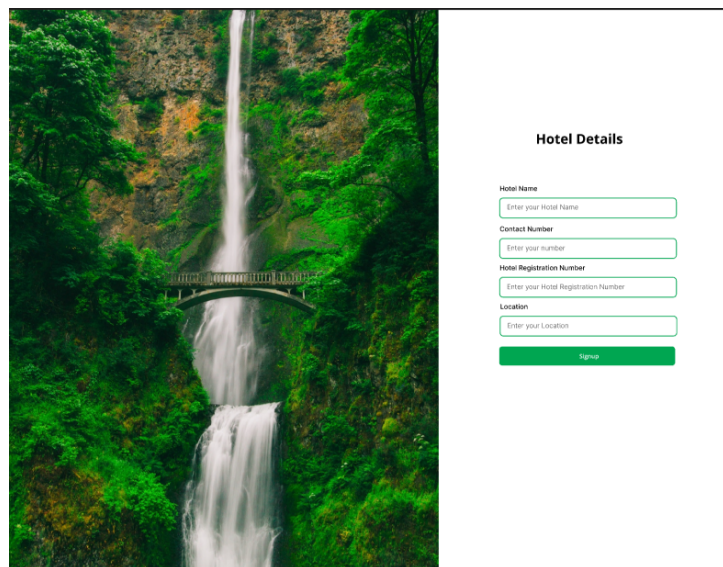


Figure 4.1: Sign up Page

This figure represents all the sign up UI of Rahzil

**Figure 4.2: Sign in Page**

This figure represents all the sign in UI of Rahzil

**Figure 4.3: Register Hotel Page**

This figure represents Hotel registration UI of Rahzil

RahZil

Home

Accommodations

Profile

Enter Your Travel Plan

Enter your Departure City

Enter your Destination City

Enter your Departure Date

Enter your Return Date

Enter number of travelers

Enter your travel budget

Submit

Figure 4.4: Home Page

This figure represents Home page UI of Rahzil

4.9 Database Design

4.9.1 ER Diagram

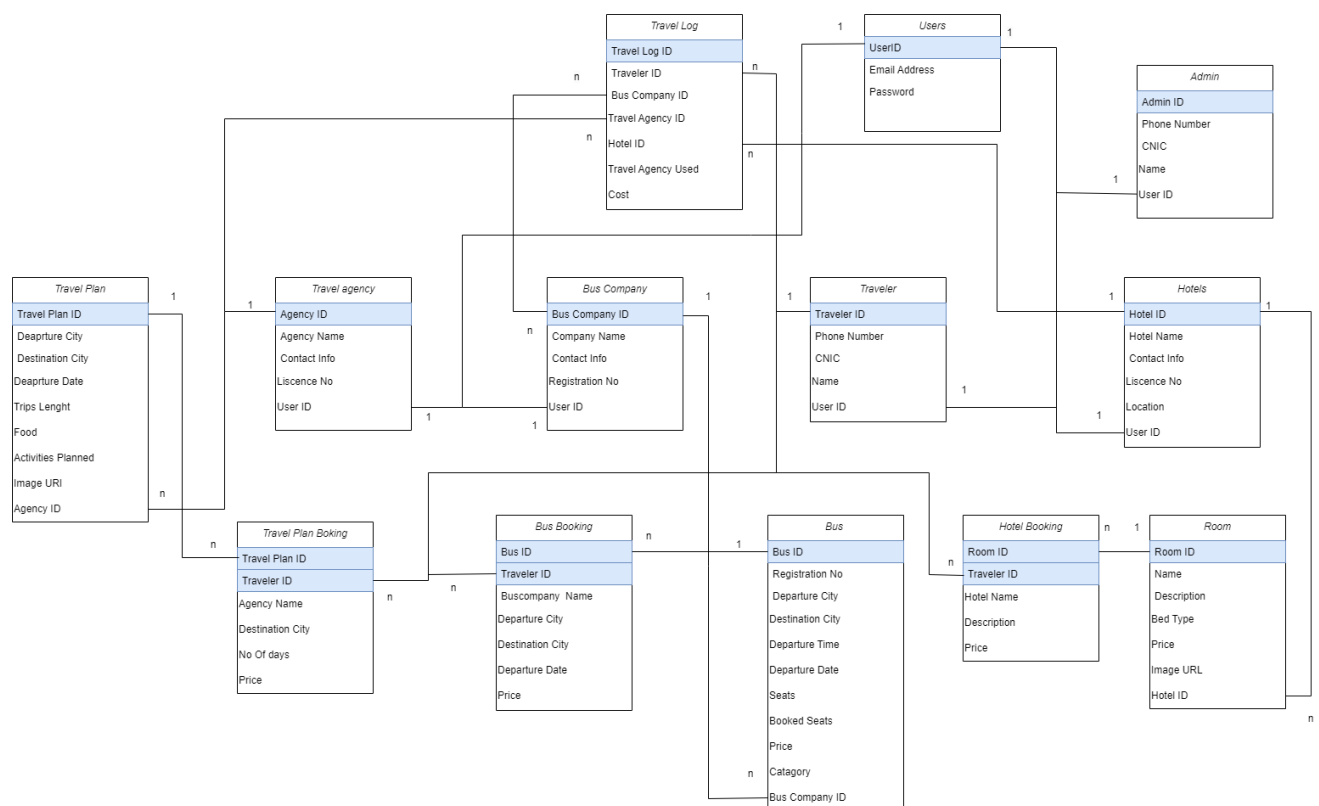


Figure 4.5: ER diagram of Rahzil

This figure represents database design for Rahzil

4.9.2 Data Dictionary

Table 4.1: Data Dictionary

This table consist of all the detail description of entities used in above ER diagram

Entity	Attribute	Data Type	Nullable	Relation To	Relation Type	Description
Admin	Admin ID	int	No			Primary Key
	Name	string	No			Full Name
	CNIC	string	No			CNIC
	Phone No	string	No			Phone No.
	User ID	int	No	Users	1 to 1	Foreign Key
User	User ID	int	No	Traveler	1 to 1	Primary Key
				Admin	1 to 1	
				Bus Company	1 to 1	
				Hotel	1 to 1	
				Travel Agency	1 to 1	
	Email Address	string	No			Email
Traveler	Traveler ID	int	No	Bus Booking	1 to n	Primary Key
				Travel Plan Booking	1 to n	
				Hotel Booking	1 to n	
				Travel Log	1 to n	
	Name	string	No			Full Name of Traveler
	CNIC	string	No			CNIC of Traveler
	Phone No	string	No			Phone No.
	User ID	int	No	Users	1 to 1	Foreign Key
Bus Company	Company ID	int	No	Bus	1 to n	Primary key
				Travel Log	1 to n	
	Bus Company Name	string	No			Name of Bus Company
	Contact No	string	No			Contact No of Compnay
	Registration No	string	No			Registration No of Bus Company
Hotels	Hotel ID	int	No	Room	1 to n	Primary key
				Travel Log	1 to n	
	Hotel Name	string	No			Name of Hotel
	Contact No	string	No			Contact No of Compnay
	Registration No	string	No			Registration No of Hotel
	Location	string	No			Location of Hotel

Travel Agency	Agency ID	int	No	Travel Plan Travel Log	1 to n 1 to n	Primary key
	Agency Name	string	No			Name of Travel Agency
	Contact No	string	No			Contact No of Travel Agency
	Registration No	string	No			Registration No of Travel Agency
	User ID	int	No			Foreign Key
Bus	Bus ID	int	No	Bus Booking	1 to n	Primary Key
	Departure Date	date	No			Date on which bus will leave
	Seats	int	No			Total No of seats in the bus
	Booked seats	int	No			No of seats booked
Room	Room ID	int	No	Hotel Book- ing	1 to n	Primary Key
	Name	string	No			Name of the Room
	Description	string	No			Details about the room
	Bed Type	string	No			Type of bed present in room
	Image Url	string	No			Image link of room
Travel Plan	Travel Plan ID	int	No	Travel Plan Booking	1 to n	Primary Key
	Departure City	string	No			Name of city from which trip will start
	Destination City	string	No			Name of Destination
	Departure Date	date	No			Date on which plan will start
	Trip lenght	int	No			No of Days of trip
	Activities Planned	string	No			Name of entertainment Activities planned for travellers
	Image Url	string	No			Image of Destination
	Agency ID	int	No			Foreign Key

Bus Booking	Bus ID	int	No			Primary Key
	Bus Company Name	string	No			Name of Bus Company
	Departure City	string	No			Name of city from which bus is departing
	Destination City	string	No			Name of city to which bus is arriving
	Departure Date	date	No			Date on which bus will leave
Hotel Booking	Price	int	No			Price of Bus Ticket
	Price	int	No			Price of Room
Travel Log	Travel Log ID	int	No			Primary Key
	Traveler ID	int	No			Foreign Key
	Travel Agency Used	string	No			YES if used, NO if not used
	Bus Company ID	int	Yes			Bus company used
	Travel Agency ID	int	Yes			Name of Travel Agency whose Travel Plan is used
	Cost	int	No			Total Cost of trip

4.10 Risk Analysis

Risk analysis is a vital aspect of project planning, identifying potential challenges and uncertainties that may affect the project's progress. RahZil's risk analysis assesses various factors to proactively mitigate potential issues, ensuring a smooth development process and successful project delivery.

- **Hardware Failures:** Although RahZil relies on minimal hardware requirements, unexpected hardware failures or malfunctions can disrupt service availability. To mitigate this risk, regular hardware maintenance and backup systems are implemented, ensuring rapid recovery in the event of hardware failures.
- **Software Compatibility:** It's critical that RahZil works with a variety of web browsers. Difference in browsers rendering might cause incompatibilities.
- **Data security:** RahZil uses the strong encryption protocol provided by HTTP, and sticks to indus-

try best practices to protect user information and prevent data breaches.

- Third-Party Integrations: RahZil depends on some platforms to provide features like weather forecasts and its functionality may be impacted by outages or other interruptions in these services.
- User Experience: Intuitive design and easy to user interface can help improve user experience. RahZil aims to provide users with the highest quality of usability, desirability and usefulness.

4.11 Conclusion

This section covered in detail the technical requirements of RahZil. The SRS provides a clear strategy for the development team to ensure RahZil is seen under the same light and its objectives are known to all. Risks have also been anticipated and measures have been established to help reduce the impact of uncertain situations that might befall RahZil. This can help in guiding the development process and creating a user-friendly platform with greatest possible usability.

Chapter 5 High-Level and Low-Level Design

5.1 System Overview

By displaying hotel and bus tickets as well as travel schedules from businesses within the user's price range, the application that is to be developed enables travel. Following are the details of modules that our system has:

5.1.1 Reserve Buses

Our software is designed to make bus booking as simple and economical as possible. Our technology will allow customers to select the best car for their budget and trip requirements. Tours may be simply organised and timed accordingly.

5.1.2 Reserve Hotels

Our app also allows users to book hotel rooms. The software provides users with several options for planning hotels and budgets. The user may select a room based on the sort of bed they want. Finding the ideal hotel without going over budget is easier and faster with our app.

5.1.3 Reserve Travel Plans

Our application helps our user to reserve travel plan which are offered by travel companies. The user gives the website all the required information and the app provides him with the best travel plan. User can look at all the details and then book the travel plan if the user likes it.

5.1.4 ToDo List And Weather Forecast

To help users plan their trips more effectively, our software also offers the feature of allowing users to create ToDo lists for each trip they take. Users may easily use our module to construct a ToDo list for their vacation in order to keep track of all the fun things they intend to undertake and to put down crucial planning points. Furthermore, our app has a function that allows users to view the weather conditions of the location they intend to visit so that they may plan properly.

5.2 Design Considerations

Following section discusses the design considerations that need to be discussed in order to make our app function properly.

5.2.1 Assumptions and Dependencies

The assumptions and dependencies related with our web application that must be handled are listed below.

5.2.1.1 Operating system

To access the web app, users are required to have the updated version of Google Chrome, Internet Explorer, Mozilla Firefox, or Safari installed on their machines.

5.2.1.2 End-User Characteristics

- The person using it will have a consistent web connection.
- The person who uses it must have a device with the most recent browser version.
- We assume that user will know how to run the application.
- The users should have google maps and their location turned on to access location services.

5.2.2 General Constraints

Our application aims to provide comfort to tourists so they can easily plan their trips and manage their budget. However, these are some constraints that can impact the performance of our application. These constraints are listed below:

5.2.2.1 Hardware and software environment

- No internet connection.
- Users are having difficulty accessing the website due to too much internet traffic.
- Browser version not supported.
- The website is hacked and shut down.

5.2.2.2 End-User environment

- Users don't have knowledge about the internet.
- Users don't have the knowledge of how to use a web application.
- Users don't know how to use Google Maps.

5.2.2.3 Availability or volatility of resources

- Weak Internet Connection.
- Defective Device.

5.2.2.4 Interoperability requirements

- A good connection between the connected user and the database.
- Application is compatible with various browsers, operating systems and devices.
- Application supports integrating with third-party applications and services through Application Programming Interfaces(APIs).

5.2.2.5 Security Requirements

- Username and passwords are stored in database in encrypted form.
- A token is generated for each user during a session.

5.2.2.6 Interface/protocol environments

- An easy-to-use interface for navigating the system.
- An easy-to-use interface to input and display data.

5.2.2.7 Performance Requirements

- The application must be able to deal with a large number of users at one time.
- The application must be available for atleast 95 percent of time over a 24 hour period.
- The application takes at most 5 seconds to retrieve data from database.

5.2.3 Development Methods

Scrum methodology will be our development method. Scrum is an easy-to-use, adaptable framework for leading and finishing complicated projects. Work is broken down into fixed-length iterations called sprints in Scrum, with frequent review and changes to produce useful results. A daily sprint meeting is scheduled to discuss the work that has to be done today. Moreover, a sprint review meeting is scheduled after every sprint completion to review the progress and a sprint retrospective meeting is scheduled before the start of a new sprint to identify the sprint goals and any other changes that are required.

5.3 System Architecture

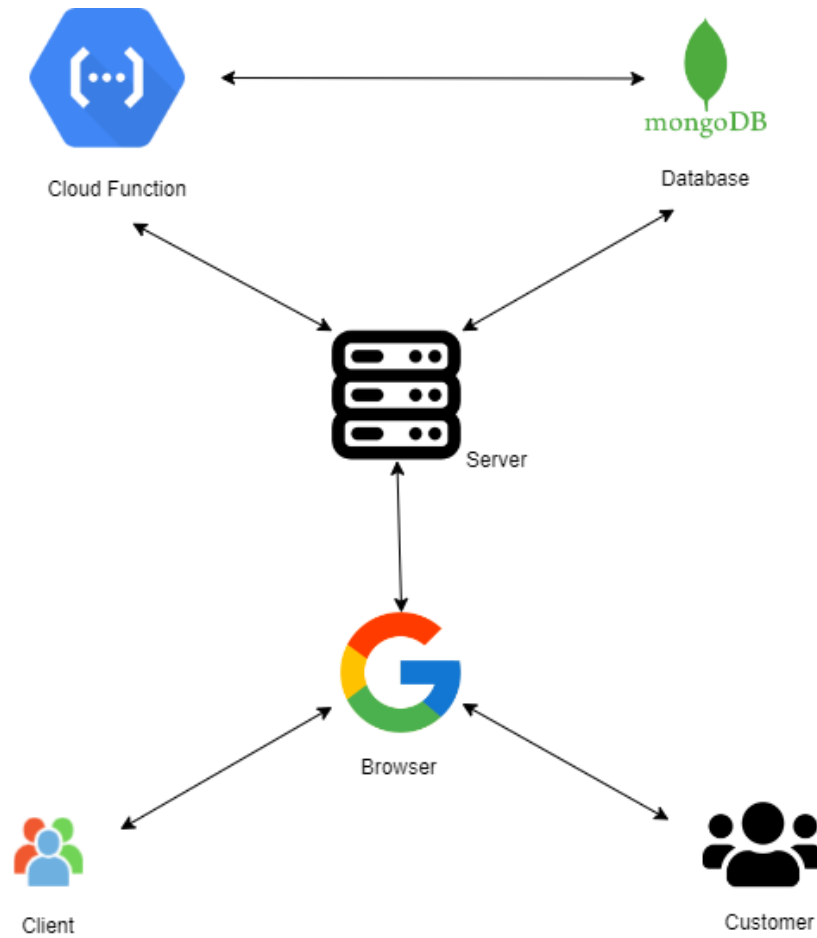


Figure 5.1: System Architect

This picture contains the system architecture of RahZil that explains the deployment and data flow in application.

The RahZil application is designed to be a travel companion for both local and international tourists who want to move across Pakistan. The system architecture is designed to support various functionalities such as bus reservations, accommodation bookings, travel guidance, budget management, and weather forecasting.

The server acts as a central point, allowing data flow between the database and external APIs. It processes requests from the client application, and perform functions like booking confirmations and data retrieval. Cloud functions are used for computationally intensive tasks, ensuring the server's performance remains optimal. External APIs provide additional data, enhancing the server's capability to deliver services to the end-users.

The flexibility of this design has made it possible for effective development and maintenance. Other designs, such a monolithic system, were taken into consideration but discarded because of their inability to scale.

5.3.1 Subsystem Architecture

Subsystems within the server are responsible for managing individual functions such as session management, user authentication, and API request processing. Together, these subsystems guarantee a flawless user experience.

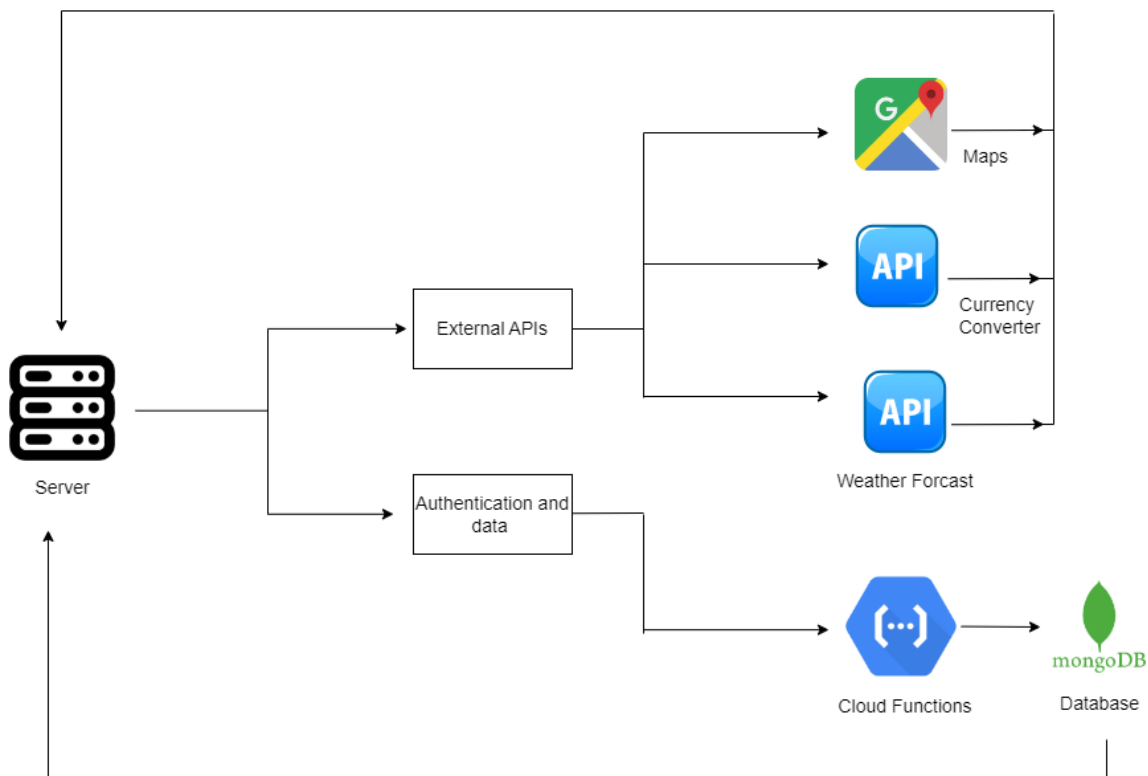


Figure 5.2: Subsystem Architect

This picture contains the subsystem architecture of RahZil that show how multiple components of app are integrated together by use of server.

Clients submit requests to the server. After that, it handles data management and authentication. It contacts the database to get or save user-related information.

The server may use external APIs, depending on the request, to retrieve additional data that the client requests. This data could include maps, currency rates, or meteorological data. To carry out particular tasks, such as processing data, doing calculations, or communicating with other services or the database, the server may also make use of cloud functions.

With the help of this architecture, the application can assign some tasks to specialized services (such as server less computing platforms or third-party APIs). This will streamline the codebase of the server and increase scalability.

5.4 Architectural Strategies

We are using Cross-Platform Compatibility and React’s component-based architecture. This will help us to write code more quickly and reuse that. The application’s cross-platform design guarantees increased accessibility and reach.

5.4.1 Integration with External APIs

The program interfaces with established, trustworthy APIs to forecast the weather and convert currencies.

5.4.2 Future Extensibility

As the platform develops, it will be simple to add new features and integrate with additional services thanks to the architecture’s focus on future extensibility.

5.4.3 Error Detection and Recovery

The system has strong error handling in place to guarantee stability and give users insightful feedback on any problems they may encounter.

5.4.4 Distributed Data and Control

For data and control, the system uses a distributed architecture, which guarantees fault tolerance and high availability.

5.4.5 Communication Mechanisms

Client-server communication uses RESTful APIs, which provide a standard protocol for data exchange.

5.4.6 Resource Management

The system architecture employs strategies for efficient resource management, including load balancing and dynamic resource allocation. These will ensure optimal performance.

5.4.7 Performance Optimization

The use of Cloud Functions and a distributed database system aims to ensure that the application remains scalable and maintains high performance, even as user numbers grow.

5.4.8 Security and Data Privacy

A strong emphasis is placed on security and data privacy. Strategies such as encryption, secure API design, and compliance with data protection regulations are being implemented.

5.5 Class Diagram

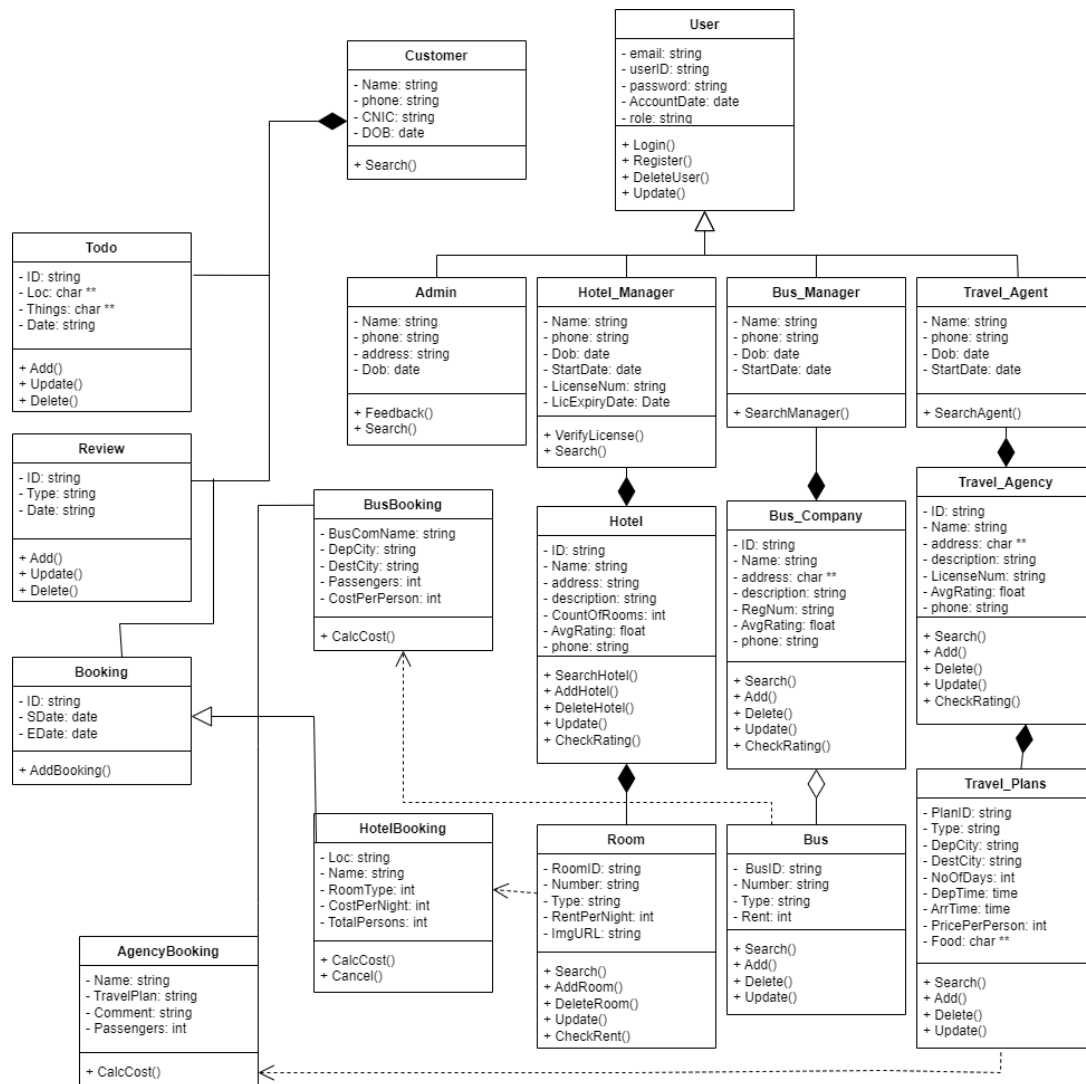


Figure 5.3: UML Class diagram

This figure represents UML Class Diagram for the functional and structural working of the Rahzil application

5.6 Sequence Diagrams

5.6.1 Authentication

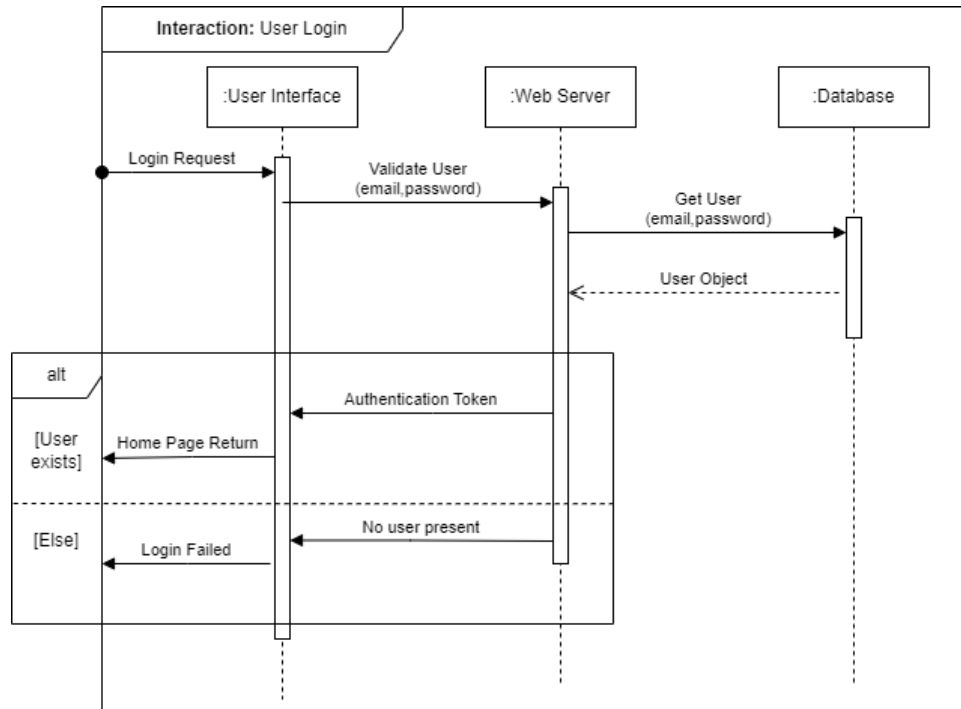


Figure 5.4: User Identification Sequence Diagram

This figure represents Sequence Diagram for client and customer identification and authorisation.

5.6.2 Book Company

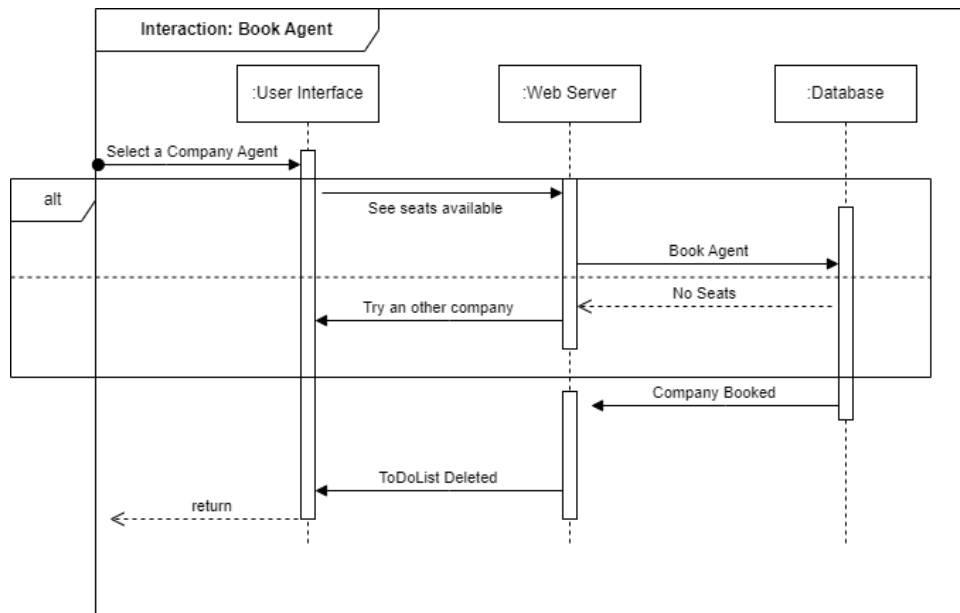


Figure 5.5: Book Company Trip

This figure represents Sequence Diagram customer to book a company of their choice and is that trip is not available.

5.6.3 Registration

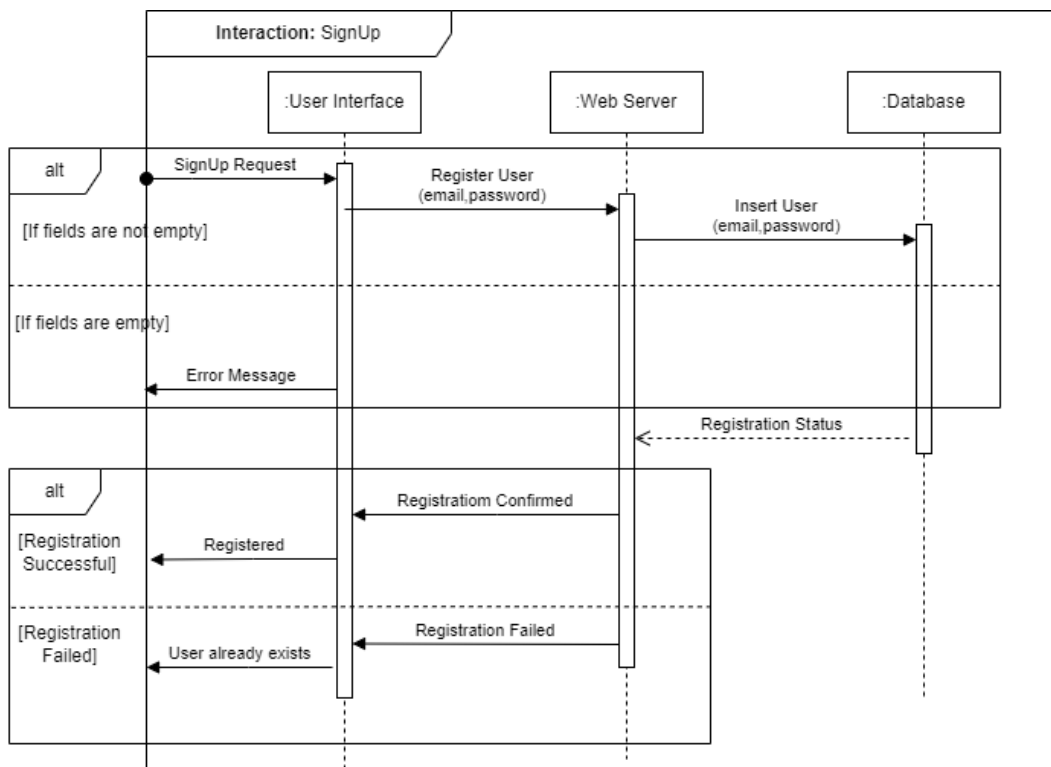


Figure 5.6: Client/Customer signup

This figure represents Sequence Diagram customer to register a company or hotel or bus for services of their choice and client for selecting a trip of their choice.

5.6.4 ToDo List

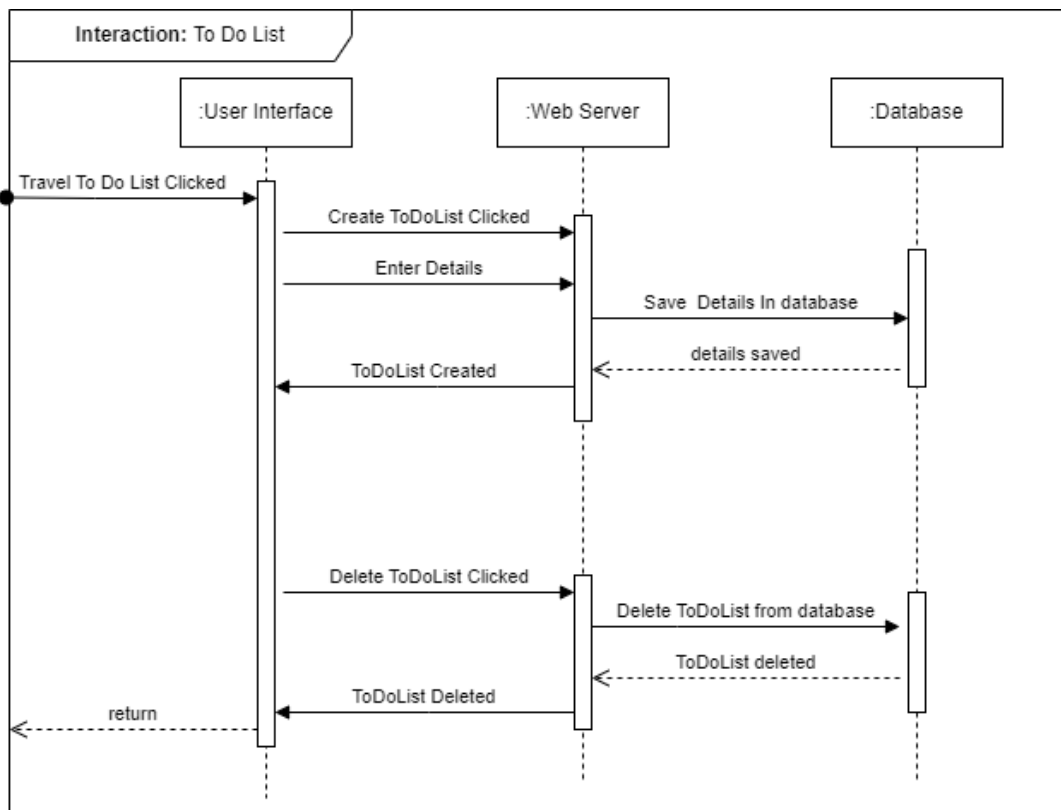


Figure 5.7: ToDo List

This figure represents Sequence Diagram for a customer that can maintain a todo list of all his or her plans for the trip.

5.7 Policies and Tactics

RahZil employs a series of guidelines based on our vision for the application to guarantee the platform's efficient operation and user's resolute satisfaction.

5.7.1 Tools

RahZil utilizes a diverse range of tools to ensure an efficient development process and a worthwhile end product. Our web application uses the React libraries to create interactive and responsive user interfaces. The component-based design improves code reusability and maintainability. For backend development, Node.js comes into play while for data management and storage, MongoDB accommodates the dynamic nature of our web application. Visual Studio Code, being a powerful yet lightweight integrated development environment, is the main tool to be used for the development phase. Along with that, we will use draw.io for diagram creation to fully make the structure of the application transparent.

5.7.2 User Policy and Security

RahZil is dedicated to protecting user privacy and guaranteeing the security of personal information. Strict compliance with data protection laws, secure authentication, and the use of secure communication channels are mandated by policies.

5.7.3 Maintenance and Improvement

RahZil is committed to innovation and ongoing development. To improve the platform's features and services, tactics include routine evaluations of user input, technology developments, and market trends. RahZil strives to continuously improve the user experience and hold a leading position in the travel utility application market by giving priority to customer feedback.

5.7.4 End User Interface

RahZil has been deliberately built as a cutting-edge, web-based application that conforms to our users' varied range of devices. RahZil is not just accessible but also designed for reading and interaction on a variety of devices, including mobile phones, computers, and tablets, as part of our dedication to provide an outstanding user experience.

5.7.5 Testing the System

To test the application functionality, black box testing is our foremost choice. Different inputs will be generated to compare the expected output and the actual output. The fundamental benefit of black box testing is its emphasis on the user perspective, which ensures that the software satisfies the user's requirements and expectations.

Chapter 6 Implementation and Test Cases

6.1 Implementation

We have developed a prototype to show the fundamental functions of our application. It is a web application built using React JS for the frontend and NodeJS for the backend with mongoDB.

First, it takes you to the login page. If the user is new to the app, he/she can click on signup, and it will take you to the signup page. The user enters his/her email and password, and depending on the role he/she selected, it will ask the user for some additional information. After a successful login, it takes the user to the bus dashboard, where they can enter details and see the bus schedule.

If the user wants to see accommodation details or travel plans, he/she can navigate to the respective page, enter the details, and see the services offered. Lastly, if the user is the manager of a bus company, hotel, or travel agency, they can add details according to their respective services for users to acquire. We have successfully integrated hotel and bus services according to user specifications. This integration includes displaying options and maintaining user history. The data presented on web pages is fetched from the database and seamlessly integrated with both the backend and frontend pages.

6.2 Test case Design and description

All the test cases performed to ensure proper functionality of our system during the testing phase are elaborated below.

Table 6.1: Sign Up*This is the sign up process of our application*

Sign Up			
UC-01			
Test Case ID:	1	QA Test Engineer:	Abu Bakar Saddique
Test case Version:	1	Reviewed By:	Umama Najam
Test Date:	28-04-2024	Use Case Reference(s):	4.6.1: Sign Up
Revision History:	None		
Objective:	To check if admin is able to create a new users account		
Product/Ver/ Module:	Signup module of website		
Environment:	Website is running on any web browser and internet is connected.		
Assumptions:	None		
Pre-Requisite:	The user is logged in the system and has the rights to create new user.		
Step No.	Execution description	Procedure result	
1	User clicks the create new user in the navigation bar.	System takes the user to create new user screen.	
2	User enters all the required details and clicks submit.	System creates new user in the back-end.	
Comments: The test case is passed. Our system is working according to our need.			
Passed			

Table 6.2: Sign in*This is the login process of our application*

Sign in			
UC-02			
Test Case ID:	2	QA Test Engineer:	Daud Abdul Rehman
Test case Version:	1	Reviewed By:	Abu Bakar Saddique
Test Date:	28-04-2024	Use Case Reference(s):	4.6.2: Sign In
Revision History:	None		
Objective:	To check if user is successfully able to login to the application		
Product/Ver/Module:	Login module of website		
Environment:	Website is running on any web browser and internet is connected.		
Assumptions:	Login button is visible.		
Pre-Requisite:	The user is a valid user with credentials stored in the database.		
Step No.	Execution description	Procedure result	
1	User enters their credentials, i.e., their username and password, and clicks login.	System takes the user to the home screen.	
Comments: The test case is passed. Our system is working according to our need.			
Passed			

Table 6.3: Signout*This is the logout process of our application*

Signout			
UC-03			
Test Case ID:	3	QA Test Engineer:	Daud Abdul Rehman
Test case Version:	1	Reviewed By:	Abu Bakar Saddique
Test Date:	28-04-2024	Use Case Reference(s):	4.6.3: Sign out
Revision History:	None		
Objective:	To check if user is successfully able to logout out of the system		
Product/Ver/Module:	Sign out module of website		
Environment:	Website is running on any web browser and internet is connected.		
Assumptions:	Logout button is visible.		
Pre-Requisite:	The user is logged in the system.		
Step No.	Execution description	Procedure result	
1	User clicks the logout button	System takes the user to the login screen.	
Comments: The test case is passed. Our system is working according to our need.			
Passed			

Table 6.4: Reserve Transport*This process allows to user to book transport*

Reserve Transport			
UC-04			
Test Case ID:	4	QA Test Engineer:	Umama Najam
Test case Version:	1	Reviewed By:	Daud Abdul Rehman
Test Date:	28-04-2024	Use Case Reference(s):	4.6.4: Reserve Transport
Revision History:	None		
Objective:	To check if the user is able to book transport for travel.		
Product/Ver/Module:	Transport Booking Module of Application		
Environment:	Website is running on any web browser and internet is connected.		
Assumptions:	None		
Pre-Requisite:	The user is logged in the system and has the privileges of a Traveller.		
Step No.	Execution description	Procedure result	
1	User clicks the reserve transport option in the navigation bar.	System takes the user to reserve transport screen.	
2	User selects the preferred option by specifying details.	System processes the information, checks the availability and reserves the transport after confirmation.	
Comments: The test case is passed. Our system is working according to our need.			
Passed			

Table 6.5: Book Rooms*This process allows to user to book hotel rooms*

Reserve Room			
UC-05			
Test Case ID:	5	QA Test Engineer:	Daud Abdul Rehman
Test case Version:	1	Reviewed By:	Abu Bakar Saddique
Test Date:	20-04-2024	Use Case Reference(s):	4.6.5: Book Hotel Room
Revision History:	None		
Objective:	To check if the user is able to book a hotel room.		
Product/Ver/Module:	Room Booking Module of Application		
Environment:	Website is running on any web browser and internet is connected.		
Assumptions:	None		
Pre-Requisite:	The user is logged in the system and has the privileges of a Traveller.		
Step No.	Execution description	Procedure result	
1	User clicks the book hotel room option in the navigation bar.	System takes the user to book hotel room screen.	
2	User selects the preferred option by specifying details such as dates.	System processes the information, checks the availability and reserves the hotel room after reservation.	
Comments: The test case is passed. Our system is working according to our need.			
Passed			

Table 6.6: Book Travel Agent*This process allows to user to book travel agent packages*

Reserve Travel Agent Packages			
UC-06			
Test Case ID:	6	QA Test Engineer:	Abu Bakar Saddique
Test case Version:	1	Reviewed By:	Umama Najam
Test Date:	22-04-2024	Use Case Reference(s):	4.6.6: Book Travel Agents
Revision History:	None		
Objective:	To check if the user is able to book a travel package with an agent.		
Product/Ver/Module:	Travel Agent Module of Application		
Environment:	Website is running on any web browser and internet is connected.		
Assumptions:	None		
Pre-Requisite:	The user is logged in the system and has the privileges of a Traveller.		
Step No.	Execution description	Procedure result	
1	User clicks the book travel agent option in the navigation bar.	System takes the user to book travel agent screen.	
2	User selects the preferred option by specifying details such as dates.	System processes the information, checks the availability and reserves the travel agent package after confirmation.	
Comments: The test case is passed. Our system is working according to our need.			
Passed			

Table 6.7: Maintain Todo List*This process allows to user to maintain a todo list*

Maintain Todo List			
UC-07			
Test Case ID:	7	QA Test Engineer:	Umama Najam
Test case Version:	1	Reviewed By:	Abu Bakar Saddique
Test Date:	30-04-2024	Use Case Reference(s):	4.6.7: Maintain Todo List
Revision History:	None		
Objective:	To check if the user is able to maintain a todo list.		
Product/Ver/Module:	Todo List Module of Application		
Environment:	Website is running on any web browser and internet is connected.		
Assumptions:	None		
Pre-Requisite:	The user is logged in the system and has the privileges of a Traveller.		
Step No.	Execution description	Procedure result	
1	User clicks the todo list option in the navigation bar.	System takes the user to the todo list screen.	
2	User adds, deletes, and edits the todo list.	System updates the list accordingly.	
Comments: The test case is passed. Our system is working according to our need.			
Passed			

6.3 Test Metrics

The summary of the common ground of attributes of test case metrics is given below. 6.8.

Table 6.8: Functional Tests

This metric evaluates the performance of functional test cases

Metric	Purpose
Number of Test Cases	7
Number of Test Cases Passed	7
Number of Test Cases Failed	0
Test Case Defect Density	0
Test Case Effectiveness	100
Traceability Matrix	File is enclosed separately.

6.4 Conclusion

The major functionalities provided by our application include authentication, booking hotels, booking buses, booking trip package, maintain to-do list and weather forecasting. To ensure the proper functioning of all these components, we have designed several test cases. After running all the test cases we ensured that our application will seamlessly meet the needs of all its users.

Chapter 7 User Manual

This chapter provides guidelines for the users of our application. This manual shall include instructions for all user roles.

7.1 Traveler

This section provides guidelines for common functionalities for traveler, such as login, logout and basic navigation tips.

7.1.1 Login

To login on RahZil, follow these steps:

- Click on the login button on the home screen
- Then Enter your username and password and and click on login

7.1.2 Logout

To end your session, follow the following steps:

- Click on the User Icon on the top right of the navigation bar.
- Click the 'Logout' option from the drop-down.

7.1.3 SignUp

To signup on RahZil, follow these steps:

- Click on the login button on the home screen
- Then click on signup button
- Then Enter your username and password and select Radio button option “traveller” and click on submit
- Then enter your cnic, phone no and username and click on submit and you have successfully signed up

7.1.4 Book Bus

To book a bus on RahZil, follow these steps:

- Log in into your account

- Filled the required fields departure city, destination city, departure date, return date, no of travellers and travel budget
- Click on submit button
- The web app will show all the buses according to your needs and click on book to successfully book a bus

7.1.5 Book Hotel Room

To book a hotel room on RahZil, follow these steps:

- Log in into your account
- Filled the required fields destination city, stay budget and type of bed
- Click on submit button
- The web app will show all the hotel rooms according to your needs and click on book to successfully book a hotel room

7.1.6 Book Travel Plan

To book a travel plan on RahZil, follow these steps:

- Log in into your account
- Click on Agents plans on Navbar
- Filled the required fields departure city, departure date, destination city and destination date
- Click on submit button
- The web app will show all the hotel rooms according to your needs and click on book to successfully book a travel plan

7.2 Bus Company

Following are the steps to follow while using the bus company module:

7.2.1 Login

To login on RahZil, follow these steps:

- Click on the login button on the home screen

- Then Enter your username and password and click on login

7.2.2 SignUp

To signup on RahZil, follow these steps:

- Click on the login button on the home screen
- Then click on signup button
- Then Enter your username and password and select Radio button option “bus” and click on submit
- Then enter your bus company name, contact no and license no and click on submit and you have successfully signed up

7.2.3 View Buses

To view buses on RahZil, follow these steps:

- Log in into your account
- You can see all the details about a bus such as its departure date, time and number of booked and total seats, price, license plate, and category

7.2.4 Add Buses

To add buses on RahZil, follow these steps:

- Log in into your account
- Click on “Add Buses” on the navbar
- Fill in the required fields departure city, arrival city, departure date and time, license no, ticket price, number of seats, and category
- Click on submit button

7.2.5 View Bookings

To view bookings on RahZil, follow these steps:

- Log in into your account
- Click on “Bookings” on the navbar and you can all see the bus bookings

7.3 Hotels

Following are the steps to follow while using the hotels module:

7.3.1 Login

To login on RahZil, follow these steps:

- Click on the login button on the home screen
- Then Enter your username and password and click on login

7.3.2 SignUp

To signup on RahZil, follow these steps:

- Click on the login button on the home screen
- Then click on signup button
- Then Enter your username and password and select Radio button option “hotel” and click on submit
- Then enter your hotel name, contact no, location, and registration no and click on submit and you have successfully signed up

7.3.3 View Rooms

To view rooms on RahZil, follow these steps:

- Log in into your account
- You can see all the details about a room such as its description, price, and bed type

7.3.4 Add Rooms

To add rooms on RahZil, follow these steps:

- Log in into your account
- Click on “Add Rooms” on the navbar
- Fill in the required fields description, bed type, and price
- Upload the image of the room by clicking on upload
- Click on submit button

7.3.5 View Bookings

To view bookings on RahZil, follow these steps:

- Log in into your account
- Click on “Bookings” on the navbar and you can all see the room bookings

7.4 Travel Agency

Following are the steps to follow while using the travel agency module:

7.4.1 Login

To login on RahZil, follow these steps:

- Click on the login button on the home screen
- Then Enter your username and password and click on login

7.4.2 SignUp

To signup on RahZil, follow these steps:

- Click on the login button on the home screen
- Then click on signup button
- Then Enter your username and password and select Radio button option “agent” and click on submit
- Then enter your agency name, contact no, and license no and click on submit and you have successfully signed up

7.4.3 View Travel Plans

To view travel plans on RahZil, follow these steps:

- Log in into your account
- You can see all the details about a travel plan such as its departure and destination city, date, trip length, cost, food, and planned activities

7.4.4 Add Plans

To add plans on RahZil, follow these steps:

- Log in into your account
- Click on “Add Plans” on the navbar
- Fill in the required fields departure and destination city, date, trip length, cost, food, and planned activities
- Upload the image of the room by clicking on upload
- Click on submit button

7.4.5 View Bookings

To view bookings on RahZil, follow these steps:

- Log in into your account
- Click on “Bookings” on the navbar and you can all see the room bookings

Chapter 8 Conclusions

Development of a prototype that covers the major functionalities of our web application highlighting its utility, has brought us closer to a complete and fully functional end product. Using ReactJS and Django, our prototype enables users to signup, login and navigate to different dashboards showcasing the ability of RahZil to meet the diverse needs of its users. Our development approach has been focused on making the user interface friendly and providing them an experience with highest quality of usability, desirability and usefulness.

For now, RahZil is in a state of continuous evolution, with countless possibilities of improvement and further integration of modules. Our vision aims to completely revolutionize the process of booking and traveling within Pakistan by merging the different modules fulfilling the various needs of users into a single application. Once RahZil is fully developed as a cutting-edge travel management system, this goal will be accomplished.

Expanded the features of RahZil by refining the existing prototype. We initially aim to implement functionalities for searching, booking and reviewing buses, travel agents and hotels for the tourists while keeping in check their budget before moving towards implementation of the feature for service providers to upload their services. We utilized third party API's for the weather forecasting module. Lastly, we developed a feature for tourists to maintain a todo list which can be used as an itinerary. It is important to note that we look forward to incorporating user feedback into our final product to make RahZil a groundbreaking innovation in the tourism sector.

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